Computer Organization and Architecture

Assignment – 1

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Q1. What is Yotta?

Yotta is a data center that operates in India and provides various cloud and AI computing services. It is a component of an initiative aimed at self-reliance of India. The company Yotta was founded in 2019 in Mumbai, Maharashtra by Sunil Gupta. The organization is a part of Hiranandi Group, which is a real estate company. The parent company of Yotta is Nidar Infrastructure Ltd.

Q2. What is NPTEL?

NPTEL or National Programme on Technology Enhanced Learning is a totally free online website that provides online courses provided by IITs and IISc professors and students. People can pay some money and get a certificate of completion of the course.

Q3. What is Turing Award? Who received it recently?

The Turing Award is the "Nobel Prize of Computing." This is the most prestigious award in computer science. The prize is granted annually by the Association for Computing Machinery. In 2025, Charles Bennett and Gilles Brassard will be awarded with the Turing Award for their outstanding contribution to the field of quantum information science.

Q4. What is Raspberry Pi?

Raspberry Pi is an inexpensive and portable computer that was developed by Raspberry Pi Foundation in the UK. Its principal purpose is teaching people about programming, electronics, and computer science. It is widely used in robotics, IoT, home automation, embedded systems and educational purposes.

Q5. What should one take into consideration when choosing a programming language?

One should consider the application type when choosing a programming language. There are some important factors such as performance, ease of learning, platform compatibility, security, scalability, libraries.

Q6. What makes the GPU faster than the CPU?

The CPU consists of several powerful cores for doing calculations, however, the GPU is a number of cores to do several calculations at once

Q7. Explain CPU, GPU, and TPU.

• CPU, or Central Processing Unit, is the main processor of a computer. It handles general tasks such as running the operating system, executing programs, and managing system resources.

• GPU, or Graphics Processing Unit, is a specialized processor designed for parallel processing. It has many cores that can perform multiple calculations at the same time.

• TPU, or Tensor Processing Unit, is a processor developed by Google specifically for AI and Machine Learning. It is optimized to perform tensor operations quickly and efficiently.

Q8. What is the RAM size, word size, and operating system of my computer?

My computer is Lenovo IdeaPad Slim 3 running on Windows 11 operating system. The computer is based on 64-bit architecture; this means that the processor processes 64 bits of data at one time, which is one of the desirable features of the modern computers.

Q9. Who are the founders of Intel and AMD? The Intel Corporation was founded by Robert Noyce and Gordon Moore in July 1968, while Advanced Micro Devices Inc. was founded by Jerry Sanders along with seven other co-founders in May 1969.

Q10. What is the CPU organization of the processor used in our laptop? The laptop is equipped with 64-bit ARM-based multicore processor (Qualcomm Snapdragon X). There are several cores of CPU with cache memory for doing parallel processing.

Q11. What is the Stampeding Herd Problem?

Stampeding Herd Problem arises when there are many processes/threads waiting for the same resource; when it becomes free, they will all compete for it simultaneously.

Q12. Takeaways from the India AI Impact Summit 2026.

• AI for All: The summit highlighted the importance of making AI accessible and beneficial for everyone, not just wealthy nations.

• People, Planet, Progress: AI development should focus on improving human lives, protecting the environment, and promoting sustainable growth.

• Responsible AI: Governments and companies were urged to ensure AI systems are safe, ethical, transparent, and trustworthy.

• Global Cooperation: Countries agreed to work together on AI standards, governance, and knowledge sharing through international collaboration

Q13. What have you learnt from the movie Imitation Game?

By watching the movie Imitation Game, I have learned that the greatest inventions are usually made by those people who think outside the box. Such an innovation has been accomplished by Alan Turing trying to invent a device for decoding encoded messages.